

FACTORY-TO-CUSTOMER SHIPPING ROUTE EFFICIENCY ANALYSIS

A Data-Driven Case Study of Nassau Candy Distributor

Research Paper / Project Report

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Project: Factory-to-Customer Shipping Route Efficiency Analysis

Dataset: Nassau Candy Distributor

Analysis environment: Python, Pandas, Excel and Streamlit

This paper presents a descriptive and diagnostic analysis of shipment-route performance using the supplied Nassau Candy Distributor dataset and project specification. Factory locations are assigned from the Product–Factory correlation provided in the specification.

Submission Note

This document is an original project report prepared from the supplied dataset and project requirements. The numerical findings are calculated from the dataset used for this project. AI-assisted language editing and document formatting were used in preparing the report; the underlying dataset analysis and reported calculations are based on the supplied project data.

Abstract

This study evaluates factory-to-customer shipping route performance for Nassau Candy Distributor using 10,194 shipment/order-line records, 8,549 unique orders, 59 customer states/provinces, and 196 derived factory-to-state routes. The analysis calculates shipping lead time as the difference between Ship Date and Order Date and aggregates performance by route, factory, customer region, state/province, and ship mode. The dataset contains five factories inferred from the product-to-factory mapping in the project specification. Total recorded sales are \$141,783.63 and total gross profit is \$93,442.80.

The descriptive analysis produces an average recorded lead time of 1,320.84 days, a median of 1,274 days, and a range of 904 to 1642 days. A critical data-quality issue is present: the supplied Order Dates fall in 2024–2025 while Ship Dates fall in 2026–2030. Consequently, all records have lead times above 365 days. The results should therefore be interpreted as a diagnostic analysis of the supplied dataset rather than as validated real-world delivery-time performance.

The study concludes that route-level analytics can provide a structured way to identify high-volume routes, geographic bottlenecks, ship-mode differences, and data-quality risks. Before operational decisions are made, the source date fields should be validated and corrected where appropriate. Once validated, the same analytical framework can be reused for route benchmarking and continuous logistics monitoring.

Keywords: supply chain analytics; logistics; route efficiency; shipping lead time; data quality; factory-to-customer analysis; exploratory data analysis; Streamlit dashboard

1. Introduction

Shipping performance is a core operational dimension of distribution systems because delivery timing, route consistency, geographic coverage, and shipping-mode selection affect service reliability and resource planning. Data science and big-data analytics are widely used in supply-chain and logistics research to turn transactional data into decision-support information. A systematic review by Jahani, Jain and Ivanov identifies descriptive analysis, visualization, clustering and other analytics approaches as established methods for supply-chain and logistics applications. ■cite■turn0search0■

The present project applies a descriptive, route-level analytics framework to the Nassau Candy Distributor dataset. The project specification states that the organization lacks clarity about which factory-to-customer routes are efficient, which routes experience delays, how performance varies by region, state and ship mode, and where geographic bottlenecks exist. These requirements motivate a structured analysis based on shipment lead time and route aggregation.

2. Problem Statement

The project addresses the absence of route-level operational intelligence in the supplied Nassau Candy Distributor dataset. The required analysis focuses on identifying efficient and delayed routes, comparing performance across regions and states, evaluating ship modes, and identifying geographic bottlenecks.

2.1 Research Objectives

The study has six practical objectives: (1) clean and validate shipment records; (2) calculate shipping lead time; (3) derive factory-to-customer routes; (4) benchmark routes using average and median lead time and shipment volume; (5) compare geographic and ship-mode performance; and (6) translate the findings into data-quality and operational recommendations.

2.2 Research Questions

RQ1: Which factory-to-state routes have the lowest and highest recorded average lead times?

RQ2: How does recorded shipping performance vary across customer regions and factories?

RQ3: How do the recorded lead times differ across shipping modes?

RQ4: Which states combine relatively high shipment volume with relatively high recorded lead time?

RQ5: What data-quality limitations could affect operational interpretation?

3. Dataset and Methodology

The source workbook contains 10,194 records and 18 original fields, including Row ID, Order ID, Order Date, Ship Date, Ship Mode, Customer ID, Country/Region, City, State/Province, Postal Code, Division, Region, Product ID, Product Name, Sales, Units, Gross Profit and Cost. The project specification additionally supplies factory coordinates and a Product–Factory correlation used here to derive the factory associated with each shipment.

3.1 Factory Assignment

The raw dataset does not contain a factory field. Factory is therefore derived from Product Name using the project specification's mapping. The five factories are Lot's O' Nuts, Wicked Choccy's, Sugar Shack, Secret Factory and The Other Factory. The mapping was applied to all 15 distinct product names in the dataset; no shipment remained without a factory assignment (0 unmapped records).

3.2 Lead-Time Calculation

Shipping Lead Time is calculated as: **Ship Date – Order Date**, measured in calendar days. No negative lead times were produced in the supplied data ({negative_lead} records). The analysis retains the original dates rather than inventing replacement values.

3.3 Route Definition

A route is defined as **Factory → Customer State/Province**. For every route, the analysis calculates shipment count, average lead time, median lead time, lead-time standard deviation where available, sales and gross profit. This follows the project's requested route aggregation and benchmarking methodology.

3.4 Geographic and Ship-Mode Analysis

Performance is aggregated by customer region, state/province, factory and ship mode. Geographic bottlenecks are identified descriptively using states that combine comparatively high shipment volume and comparatively high recorded lead time. Ship-mode analysis compares the recorded lead-time distributions without claiming causal effects.

4. Data Cleaning and Validation

The analysis validated date fields, calculated lead times, checked for negative durations, standardized the route definition, and verified that every product in the source data could be linked to one of the factories specified in the project brief.

4.1 Critical Date Anomaly

The principal data-quality finding is a major temporal mismatch. The source Order Date values span 2024–2025, while Ship Date values span 2026–2030. This produces recorded lead times from 904 to 1642 days. All 10,194 records (100.0%) exceed 365 days.

This issue materially affects every route, state, region, factory and ship-mode lead-time metric. The analysis therefore does not interpret the absolute lead-time values as validated operational delivery durations. Instead, they are reported as the mathematical result of the supplied source fields and used to demonstrate the route-analysis workflow.

Recommended validation: compare the Order Date and Ship Date fields with the original transaction system, determine whether one or both fields were shifted during dataset preparation, correct the source dates if authoritative evidence exists, and rerun the analysis before using the results for operational decisions.

5. Exploratory Data Analysis

The dataset contains 8,549 unique orders across 5,044 unique customers, 5 factories and 59 states/provinces. Recorded sales total \$141,783.63, while gross profit totals \$93,442.80. The difference between sales and cost is consistent with the supplied Gross Profit field.

5.1 Factory Distribution

Factory	Shipments	Avg_Lead_Time	Sales	Gross_Profit
The Other Factory	100	1,280.3	\$1,282.25	\$152.25
Wicked Choccy's	4,152	1,321.1	\$55,352.75	\$36,053.57
Lot's O' Nuts	5,692	1,321.2	\$76,340.15	\$52,771.05
Secret Factory	217	1,321.9	\$8,587.50	\$4,344.70
Sugar Shack	33	1,340.0	\$220.98	\$121.23

Lot's O' Nuts and Wicked Choccy's account for the large majority of shipment records in the supplied dataset. This concentration is relevant when interpreting route-level results because low-volume factories produce many routes with very small samples.

5.2 Ship-Mode Performance

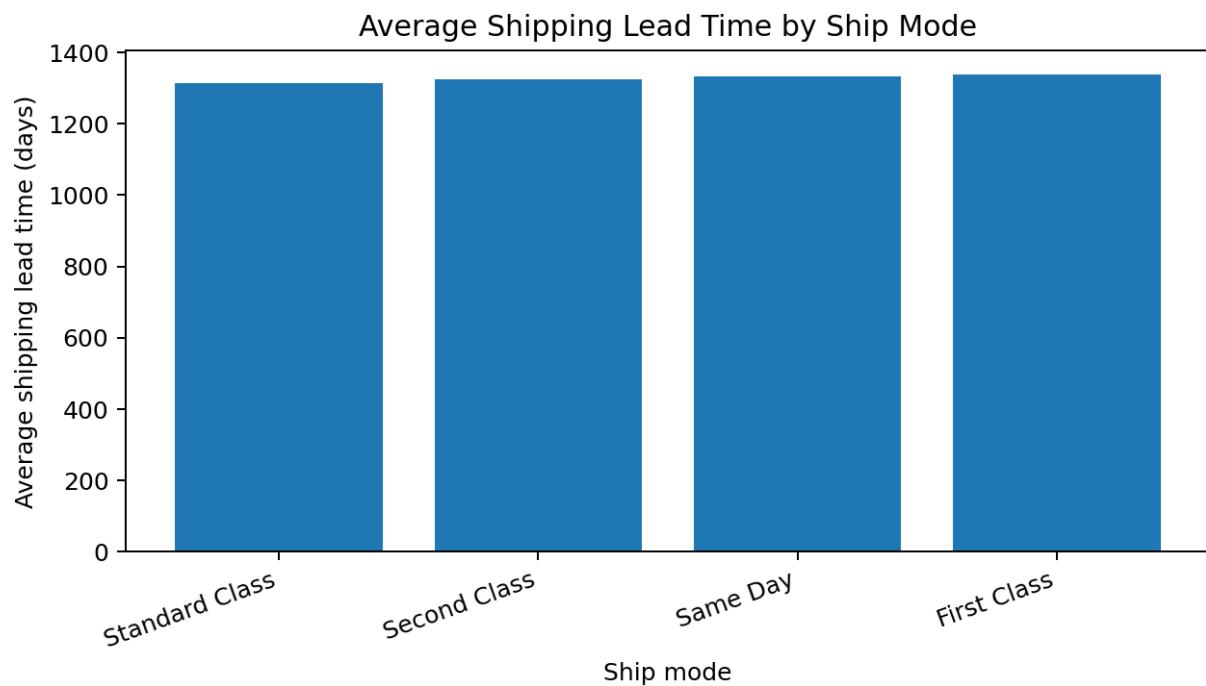


Figure 1. Recorded average shipping lead time by ship mode.

Ship Mode	Shipments	Avg_Lead_Time	Median_Lead_Time
Standard Class	6,120	1,314.3	1,274
Second Class	1,979	1,323.8	1,273
Same Day	547	1,333.4	1,269
First Class	1,548	1,338.3	1,272

The ship-mode differences are relatively small compared with the very large date-driven lead-time range. Accordingly, the observed differences should not be interpreted as evidence that one mode causes better or worse service performance.

6. Factory-to-Customer Route Analysis

The analysis identifies 196 unique factory-to-state routes. Route efficiency is evaluated primarily through average recorded lead time, with shipment count and median lead time used as supporting context. The project specification calls for ranking routes from fastest to slowest and identifying the top and bottom ten routes.

6.1 Top 10 Routes by Recorded Average Lead Time

Route	Shipments	Avg_Lead_Time	Median_Lead_Time
Secret Factory → New Mexico	2	906.0	906
Secret Factory → Nebraska	1	906.0	906
The Other Factory → Louisiana	1	907.0	907
The Other Factory → Connecticut	2	907.5	908
Secret Factory → Mississippi	1	908.0	908
Wicked Choccy's → Maine	2	908.0	908
Secret Factory → Louisiana	2	908.5	908
Secret Factory → Delaware	1	909.0	909
Secret Factory → South Carolina	1	909.0	909
Secret Factory → Minnesota	1	909.0	909

The fastest recorded routes are dominated by Secret Factory and The Other Factory routes with very small shipment counts. This illustrates why route volume should be considered alongside average lead time: a route with one or two records is less stable as a benchmark than a route supported by a large shipment sample.

6.2 Bottom 10 Routes by Recorded Average Lead Time

Route	Shipments	Avg_Lead_Time	Median_Lead_Time
Sugar Shack → New Jersey	1	1,642.0	1,642
Secret Factory → New Hampshire	1	1,641.0	1,641
Sugar Shack → Connecticut	1	1,641.0	1,641
Wicked Choccy's → West Virginia	2	1,639.0	1,639
Lot's O' Nuts → North Dakota	5	1,638.2	1,637
The Other Factory → Nevada	1	1,638.0	1,638
Sugar Shack → California	1	1,638.0	1,638
Secret Factory → Connecticut	1	1,638.0	1,638
Sugar Shack → Washington	1	1,638.0	1,638
Secret Factory → Kentucky	2	1,637.5	1,638

The slowest recorded routes include Sugar Shack → New Jersey, Secret Factory → New Hampshire, Sugar Shack → Connecticut, Wicked Choccy's → West Virginia and Lot's O' Nuts → North Dakota. Several of these routes have only one to five shipments, so the ranking is sensitive to the underlying data anomaly and small sample sizes.

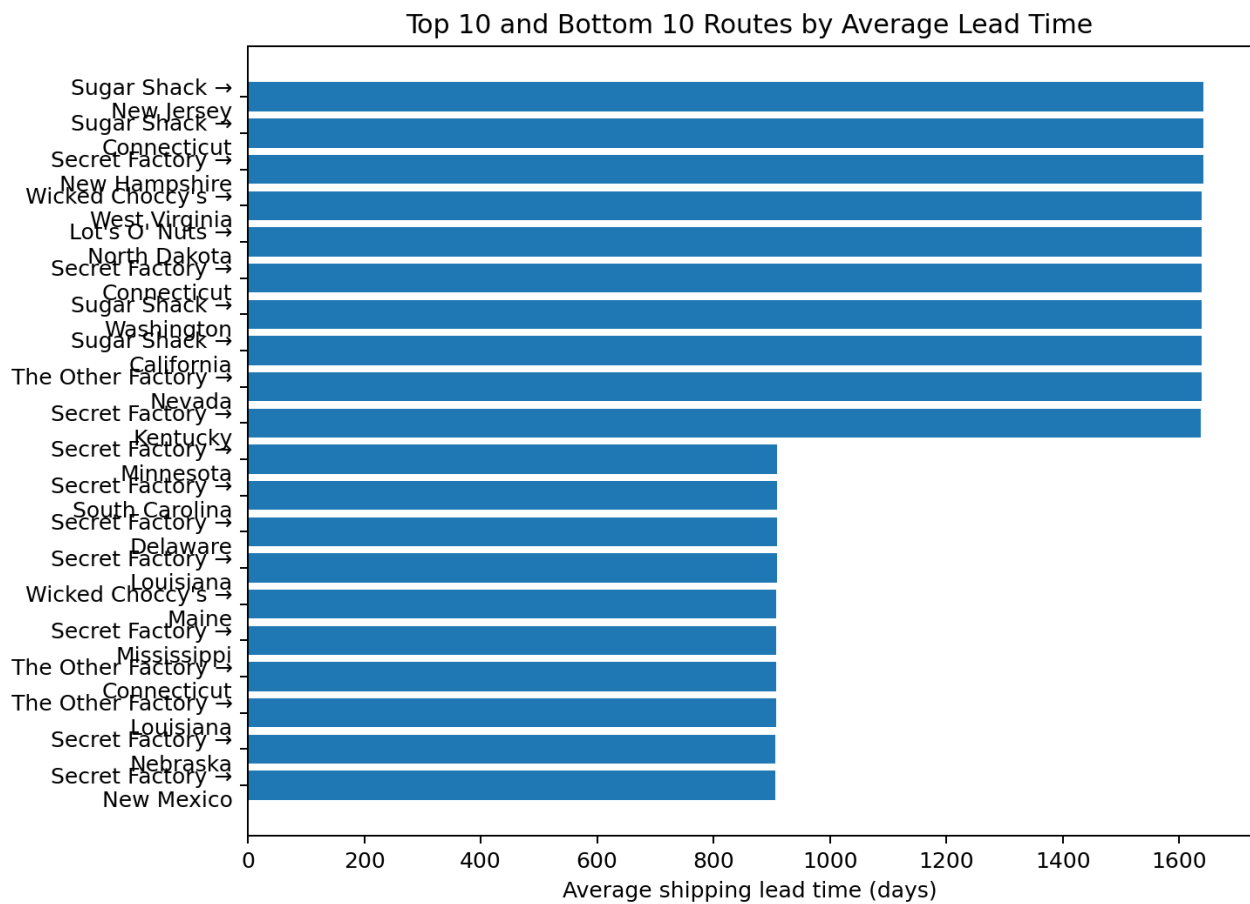


Figure 2. Top and bottom route groups based on recorded average lead time.

7. Geographic Performance and Bottleneck Analysis

7.1 Regional Performance

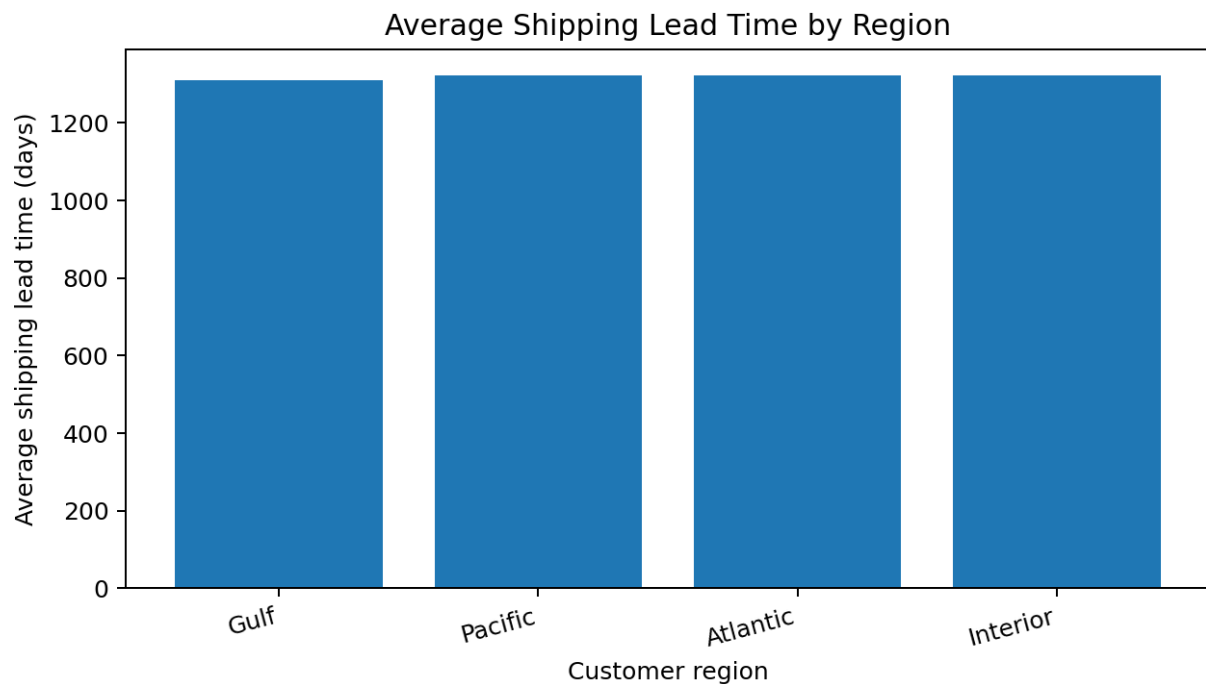


Figure 3. Recorded average shipping lead time by customer region.

Region	Shipments	Avg_Lead_Time	Median_Lead_Time
Gulf	1,620	1,311.4	1,274
Pacific	3,253	1,322.2	1,274
Atlantic	2,986	1,322.7	1,274
Interior	2,335	1,323.1	1,274

The Gulf region has the lowest recorded regional average at approximately 1,311.4 days, while the Interior region has the highest at approximately 1,323.1 days. The differences are modest relative to the overall lead-time scale and should be treated as descriptive.

7.2 State-Level Bottlenecks

State/Province	Region	Shipments	Avg_Lead_Time	Sales
Tennessee	Gulf	183	1,391.5	\$2,383.56
Indiana	Interior	149	1,381.5	\$2,002.78
Washington	Pacific	506	1,360.7	\$6,921.15
Connecticut	Atlantic	82	1,357.5	\$980.57
Maryland	Atlantic	105	1,356.8	\$1,476.99
Wisconsin	Interior	110	1,343.0	\$1,920.50
Missouri	Interior	66	1,339.9	\$882.83
Georgia	Gulf	184	1,338.6	\$2,692.84
New Jersey	Atlantic	130	1,338.3	\$1,578.35
Colorado	Pacific	182	1,337.2	\$2,544.91

The table uses a descriptive screening rule: states at or above the median shipment volume and at or above the median recorded lead time. It is intended to identify locations for investigation, not to establish operational causality.

8. Factory-Level Analysis

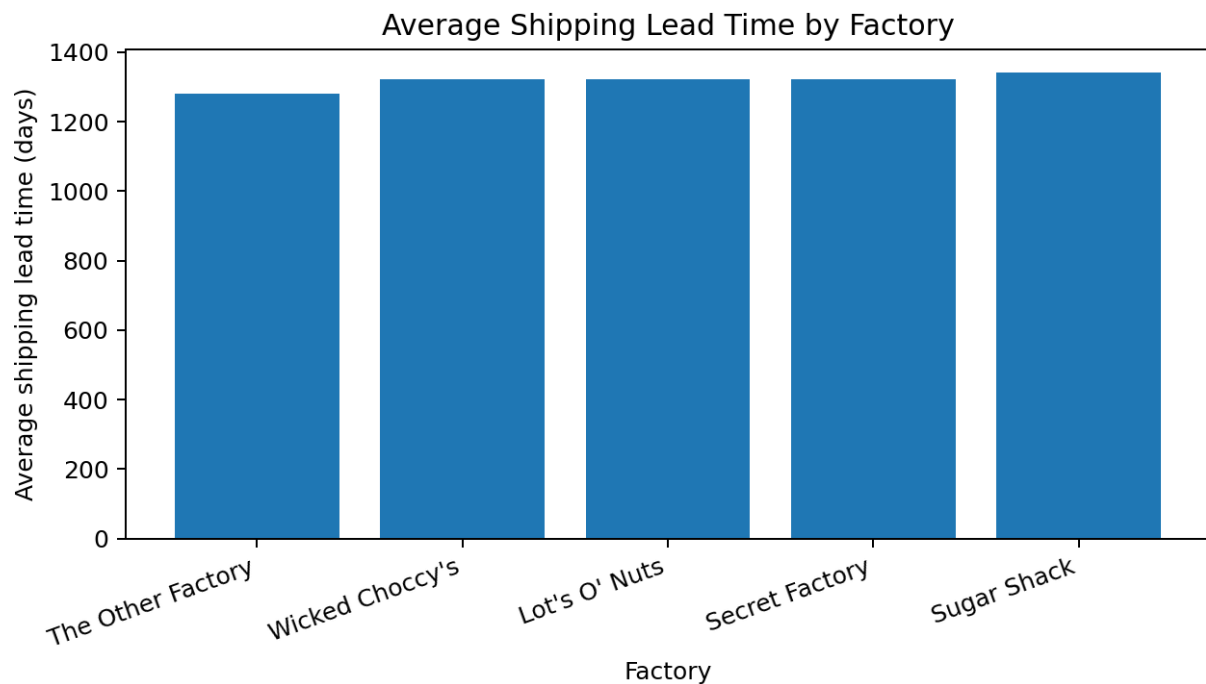


Figure 4. Recorded average shipping lead time by factory.

Factory	Shipments	Avg_Lead_Time	Median_Lead_Time	Sales
The Other Factory	100	1,280.3	1,273	\$1,282.25
Wicked Choccy's	4,152	1,321.1	1,274	\$55,352.75
Lot's O' Nuts	5,692	1,321.2	1,274	\$76,340.15
Secret Factory	217	1,321.9	1,274	\$8,587.50
Sugar Shack	33	1,340.0	1,274	\$220.98

The factory results reflect both the source-date anomaly and highly uneven shipment volumes. Lot's O' Nuts has the largest shipment count, followed by Wicked Choccy's. Consequently, these factories provide a more substantial base for aggregate descriptive statistics than the lower-volume factories.

9. Key Findings

Finding 1 — The dataset is suitable for route-analysis workflow development but not yet for validated delivery-time benchmarking. The workbook contains 10,194 records and enough geographic/product detail to construct 196 factory-to-state routes, but the date anomaly dominates the lead-time calculations.

Finding 2 — Route volume matters. The fastest and slowest route lists contain several routes with only one or two shipments. These observations should not be treated as stable route benchmarks without additional observations.

Finding 3 — Geographic differences are descriptive rather than causal. Regional and state-level averages reveal variation, but the supplied data do not contain traffic, distance, carrier, weather, vehicle or operational-event variables that would allow a causal explanation of delay.

Finding 4 — Ship-mode comparisons are constrained by the date anomaly. Differences between First Class, Same Day, Second Class and Standard Class are measurable in the supplied data, but absolute lead times should not be interpreted as real service-level durations.

Finding 5 — Data validation is itself an important project outcome. A logistics dashboard can produce polished KPIs while still being misleading if its date fields are incorrect. The project demonstrates why validation should precede operational optimization.

10. Recommendations

1. Validate the source dates first. Reconcile Order Date and Ship Date against the authoritative order-management or transportation system before using lead-time metrics for operational decisions.

2. Recalculate all KPIs after correction. Once the date issue is resolved, regenerate route averages, medians, variability, delay frequencies and ship-mode comparisons.

3. Use minimum-volume thresholds for route benchmarking. For example, a future operational dashboard could distinguish routes with fewer than five observations from statistically more stable routes.

4. Monitor high-volume/high-lead-time states. States identified by the bottleneck screening rule should receive additional review once the date data are validated.

5. Expand the dataset for root-cause analysis. Useful future fields include actual carrier, transportation distance, warehouse processing time, pickup timestamp, delivery timestamp, exception codes, promised delivery date, weather/traffic indicators and freight cost.

6. Operationalize the dashboard. The accompanying Streamlit dashboard can serve as a monitoring layer with filters for region, state, ship mode and lead-time threshold. The project specification explicitly calls for route overview, geographic analysis, ship-mode comparison and route drill-down capabilities.

11. Limitations

This study is descriptive and based solely on the supplied dataset and project specification. It does not estimate causal effects, optimize physical routes using road-network distance, or forecast future delivery performance. Factory assignment is derived from the Product–Factory mapping supplied in the project brief rather than from an original factory field in the workbook.

Most importantly, the date anomaly prevents the recorded lead-time values from being treated as validated real-world shipping durations. Therefore, this paper emphasizes the analytical framework, data-quality diagnosis and descriptive comparisons rather than claiming that the reported 900–1,600 day values represent actual logistics performance.

12. Conclusion

This project demonstrates a complete data-analytics workflow for factory-to-customer shipping route analysis using the Nassau Candy Distributor dataset. The analysis transformed 10,194 source records into 196 factory-to-state route observations and produced route, factory, regional, state and ship-mode performance views.

The most important practical result is the identification of a major source-data problem: the relationship between Order Date and Ship Date creates implausibly long recorded lead times. Detecting this issue prevents the dashboard from

being mistaken for an operational truth when its underlying dates require validation.

After the authoritative dates are corrected, the same route-level methodology can be rerun to provide meaningful benchmarks, identify persistent bottlenecks, compare shipping modes, and support continuous logistics monitoring.

References

1. Unified Mentor. *Factory-to-Customer Shipping Route Efficiency Analysis for Nassau Candy Distributor*. Project specification supplied for this analysis. The specification defines the dataset fields, route methodology, KPIs, dashboard requirements and Product–Factory correlation.
2. Jahani, H., Jain, R., & Ivanov, D. (2023). Data science and big data analytics: a systematic review of methodologies used in the supply chain and logistics research. *Annals of Operations Research*, 359, 1297–1354. <https://doi.org/10.1007/s10479-023-05390-7>
3. Data-Driven Methodology to Support Long-Lasting Logistics and Decision Making for Urban Last-Mile Operations. *Sustainability*, 13(11), 6230 (2021). The study discusses data collection, cleaning, mining and decision-support workflows for logistics operations.
4. Source dataset: *Nassau Candy Distributor*, supplied workbook used for this project analysis.

Appendix A — Core KPI Formulae

Shipping Lead Time = Ship Date – Order Date

Average Lead Time = Sum of route lead times ÷ Number of shipments on the route

Route Volume = Count of shipment/order-line records assigned to the route

Gross Profit = Sales – Cost, consistent with the project specification

Route Definition = Factory → Customer State/Province. Factory is derived from Product Name using the Product–Factory correlation supplied in the project specification.

Appendix B — Reproducibility

The calculations can be reproduced using Python/Pandas from the supplied Excel workbook. The Streamlit dashboard uses the resulting analysis workbook and provides filters and route-level tables. The Excel and dashboard files accompanying this paper should be retained with the submission package.